Cross-disciplinary applications of the Toulbar2 solver

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Abstract / Introduction

Combinatorial problem solving

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Bioinformatics & Life Sciences

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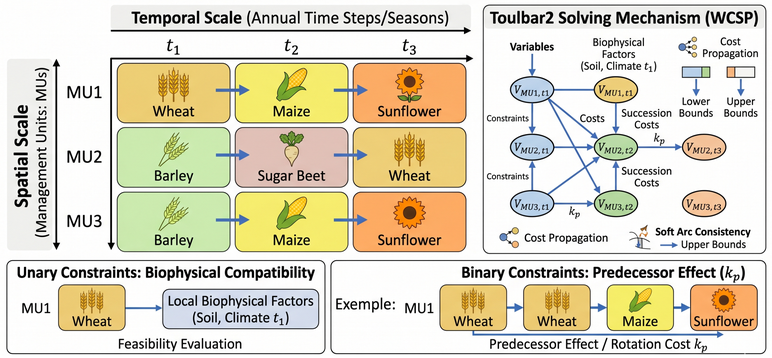
Recherche Opérationnelle & Planification

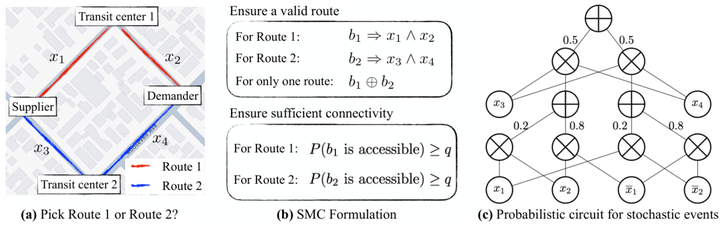
Le formalisme $WCSP$ performe dans la modélisation d'environnements complexes en combinant des règles strictes (contraintes dures) et l'optimisation de critères de qualité (contraintes souples).

Figure 1: Spatio-temporal Agricultural Modeling



Artificial Intelligence & Machine Learning

Toulbar2 struggles with real world applications since exhaustive search creates computational bottlenecks on large instances. We can integrate machine learning, introducing new ways to improve the solution quality. For example, algorithms like KOCO SMC handle large instances by tracking probability bounds during the inference process. This early evaluation prevents the solver from performing exhaustive enumerations.



(a) Pick Route 1 or Route 2? **(b)** SMC Formulation **(c)** Probabilistic circuit for stochastic events

Fig. 1: Example of SMC: Robust Supply Chain with Different Network Size.

Deep learning models can also transform raw sensory data directly into Graphical Models. This integration merges the perception of neural networks with the exact reasoning of constraint solvers to increase overall flexibility. For example, a neural network can translate unmapped protein structures or raw images into cost matrices, allowing Toulbar2 to focus exclusively on executing the exact discrete optimization.

Conclusion

Brief summary of the main takeaway from your literature review. What is the logical next step for your introductory research project?

References

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